

# FMN Sales Analytics Platform

Building a production-style data pipeline using Python, DuckDB, dbt, Airflow, and Docker

James Essiet · 6th of July, 2026





# The Business Challenge

Sales data across multiple business entities is fragmented, inconsistent, and manually processed — limiting reliable decision-making at scale.



## Automate ETL

Eliminate manual data handling with a repeatable, scheduled pipeline



## Improve Data Quality

Enforce validation rules at every stage of the data flow



## Analytics-Ready Datasets

Structure raw data into clean, queryable models for business use



## Enable Decisions

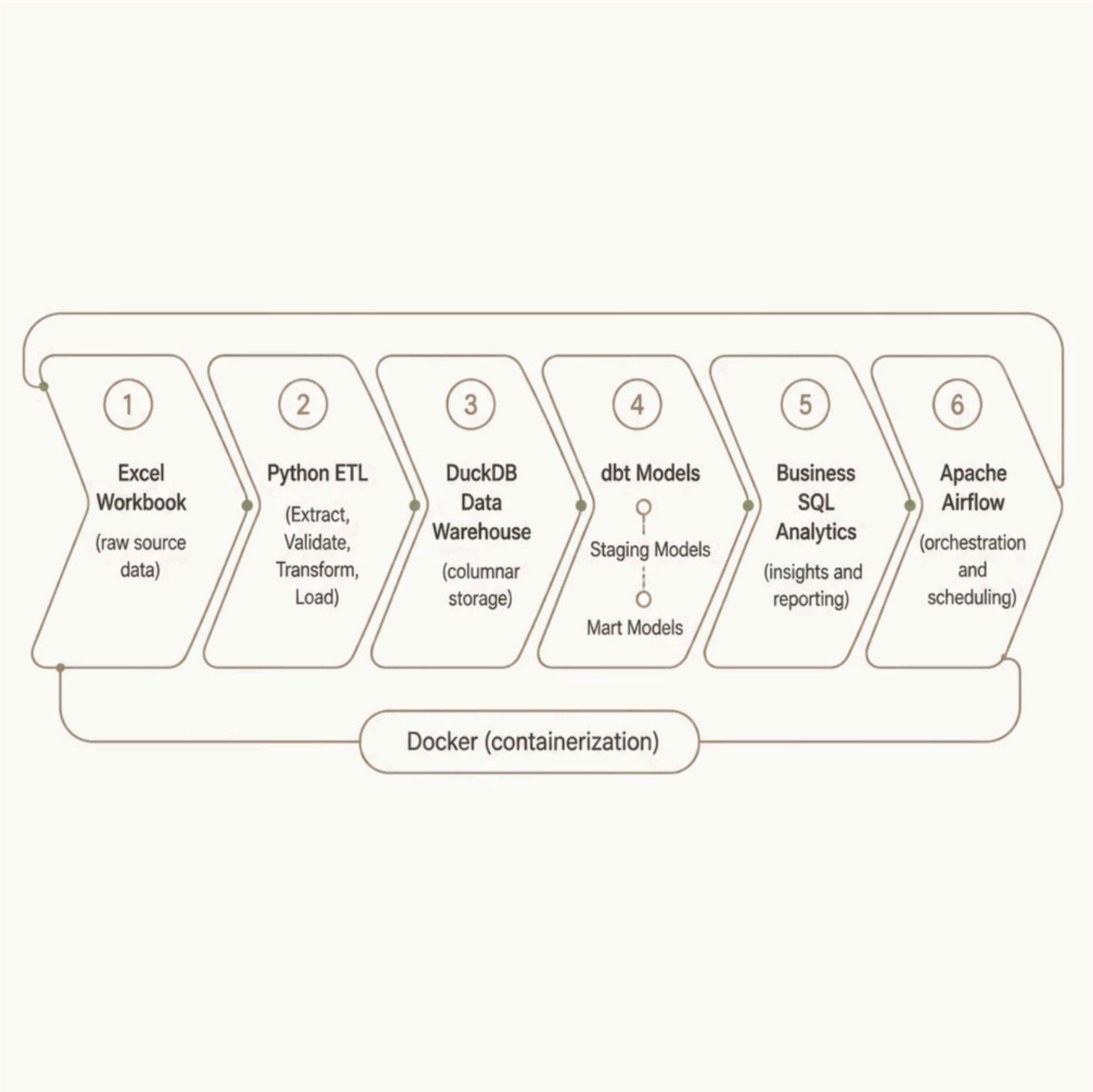
Surface actionable insights that drive revenue and operational efficiency





# Solution Architecture

A modern, containerized data stack — from raw Excel input to business-ready SQL analytics — orchestrated end-to-end with automation.



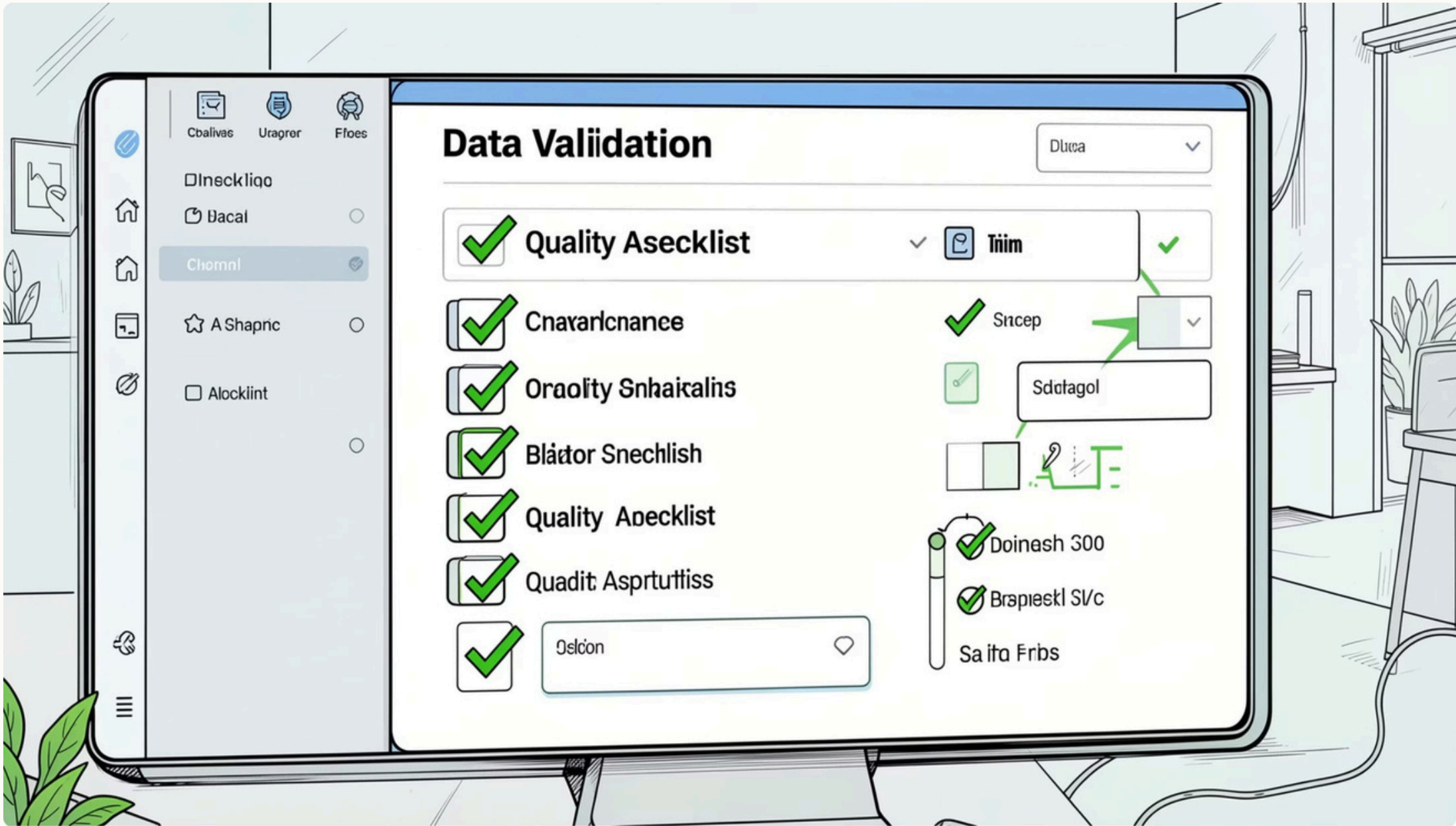
## Technology Stack

|                                                                  |                                                                            |
|------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>Python</b><br>Extract, validate, transform, and load raw data | <b>DuckDB</b><br>Lightweight, in-process analytical warehouse              |
| <b>dbt</b><br>Staging and mart model transformations             | <b>Airflow + Docker</b><br>Orchestration and reproducible containerization |

# Data Quality Framework

A comprehensive validation layer ensures only clean, trustworthy data enters the warehouse — catching issues before they reach analysts.

- Schema Validation  
Required tables and columns verified at ingestion
- Primary Key Checks  
Uniqueness enforced on all key identifiers
- Duplicate Detection  
Row-level duplicates flagged and quarantined
- Accepted Values  
Domain constraints validated against allowed enums
- Positive Numeric Validation  
Revenue, quantity, and price fields checked for non-negative values
- Foreign-Key Relationships  
Referential integrity enforced across related tables



# Analytics Engineering with dbt

dbt transforms raw warehouse tables into a structured, tested, and documented semantic layer — enabling confident self-service analytics.



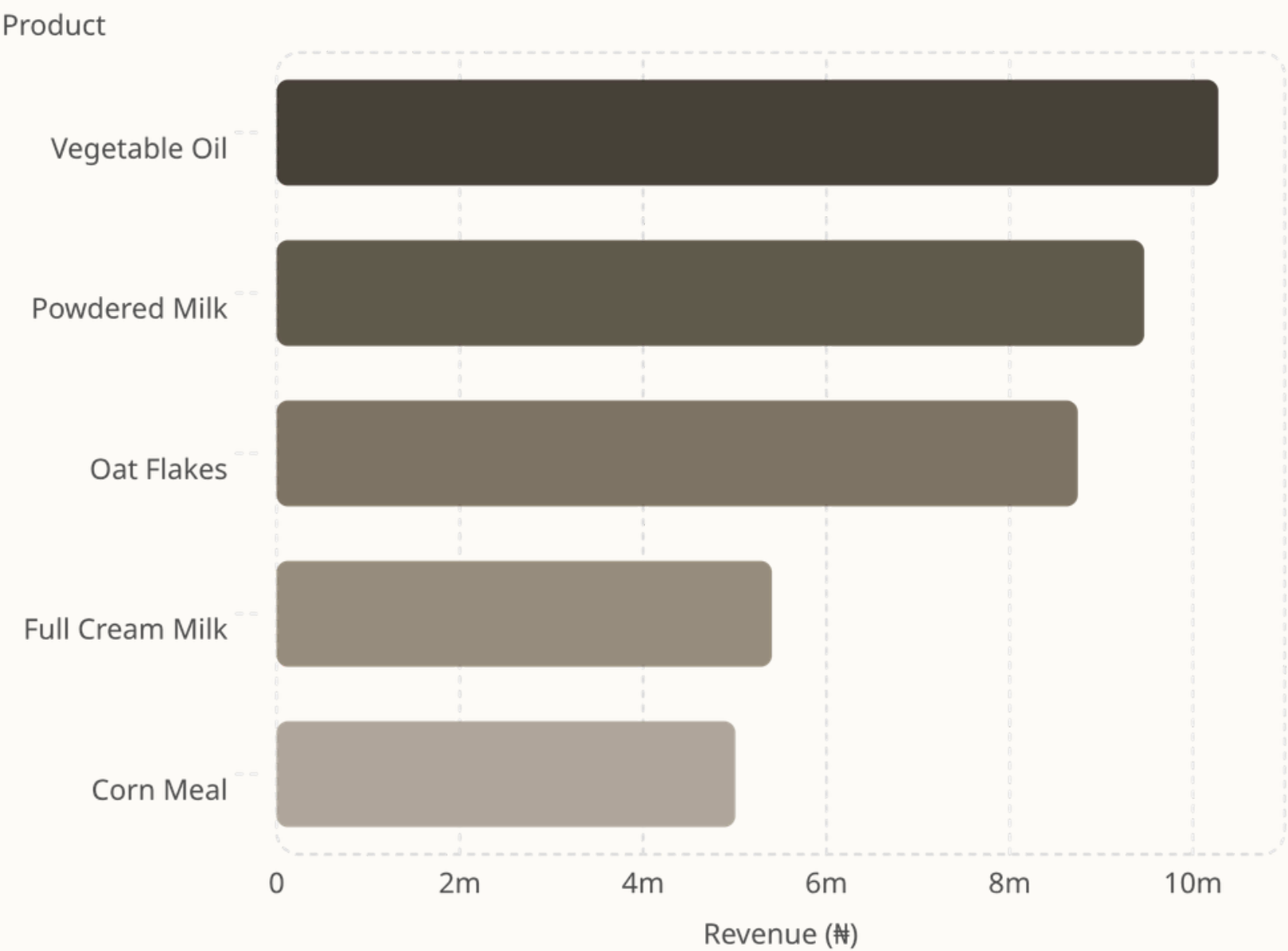
Each layer builds on the previous, creating a clear lineage from source to insight.

|                                                                            |                                                                                |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>schema.yml</b><br>Centralized documentation and metadata for all models | <b>not_null &amp; unique</b><br>Column-level constraints enforced as dbt tests |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------|

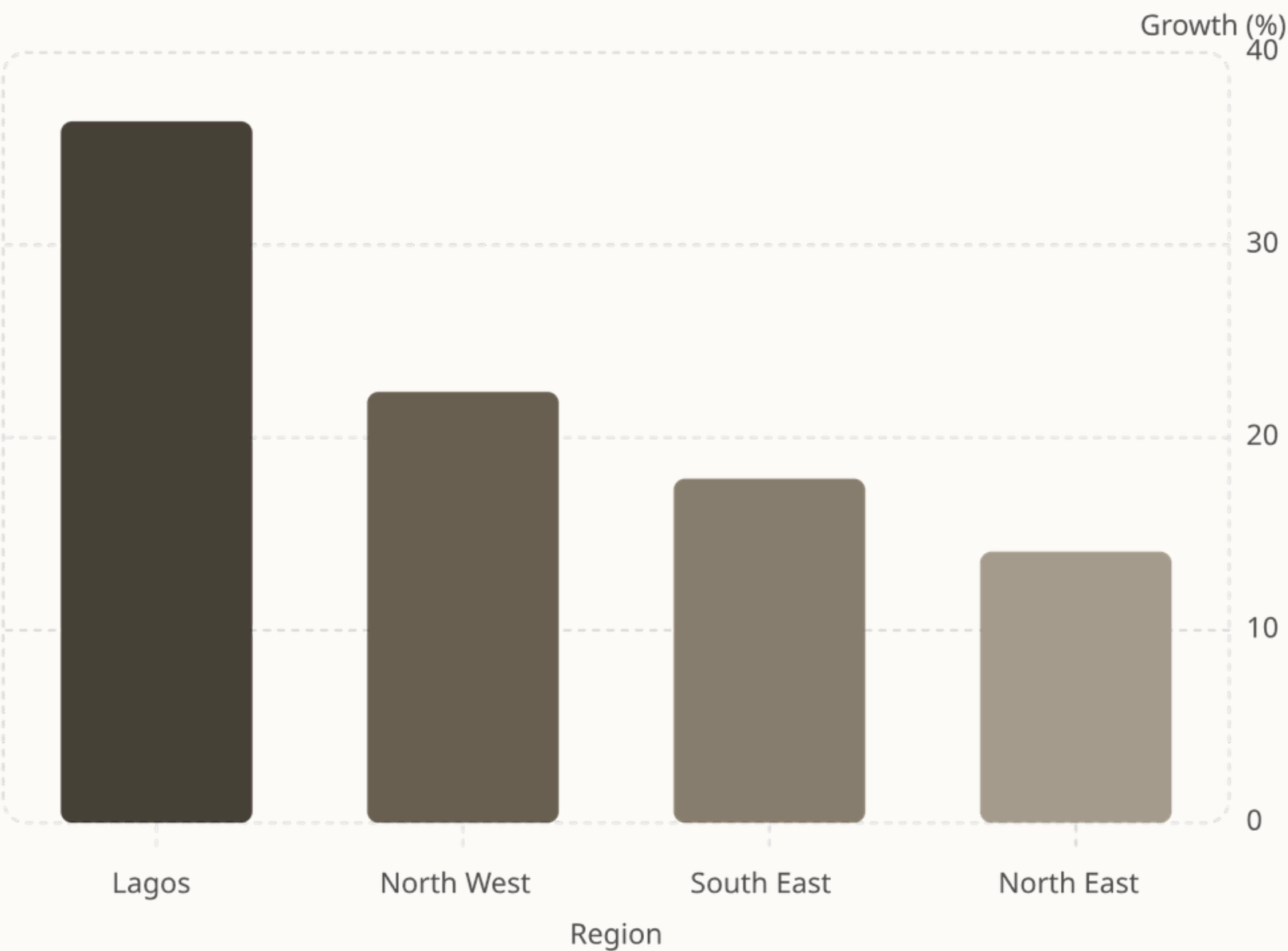
# SQL Business Insights

Direct SQL queries against the mart layer reveal clear revenue patterns and regional momentum across Q3 2025.

Top Products by Revenue (2025)



Highest MoM Revenue Growth — Q3 2025

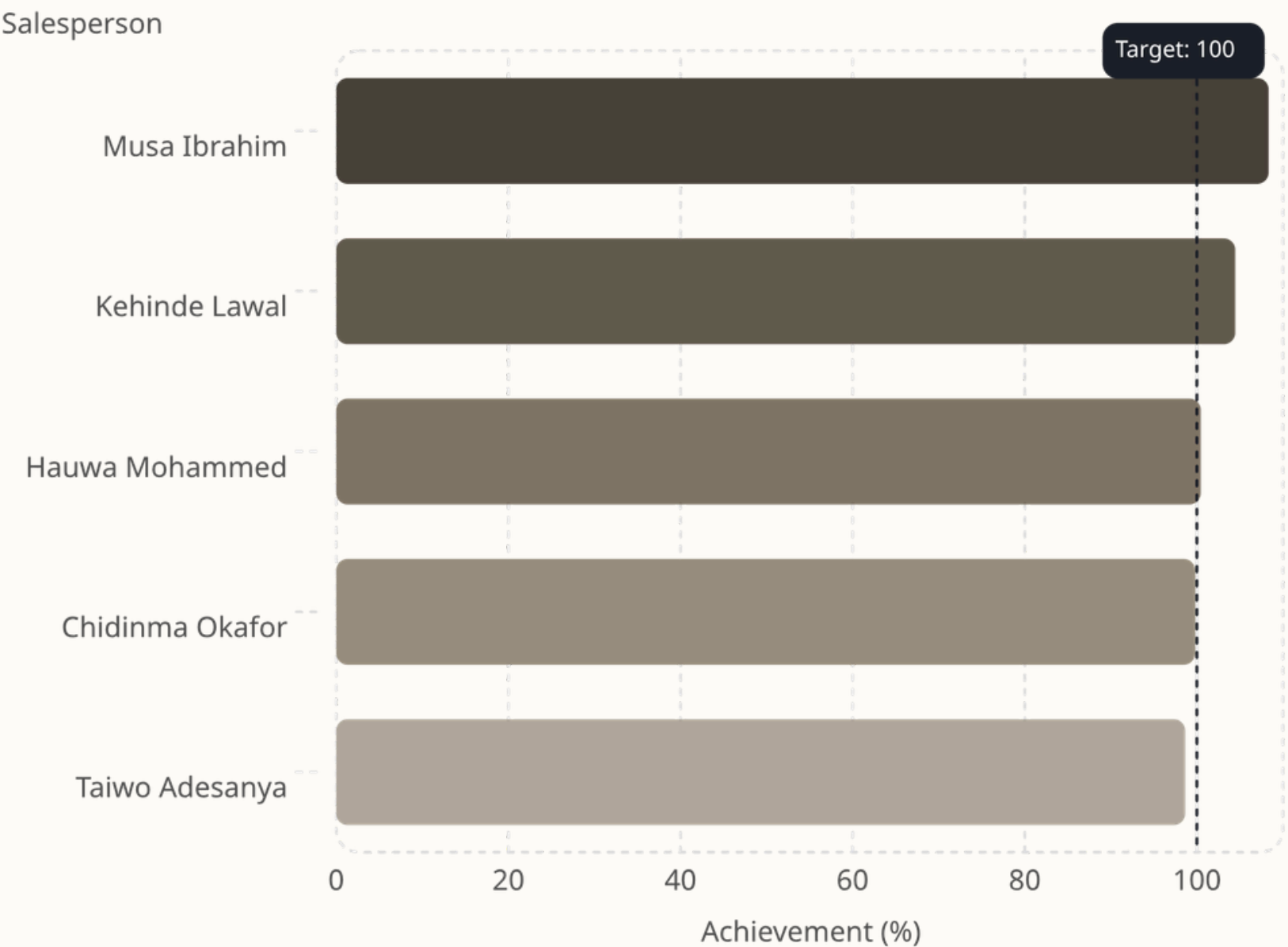


**Lagos** leads with **36.42% MoM growth** — the strongest signal for commercial investment.

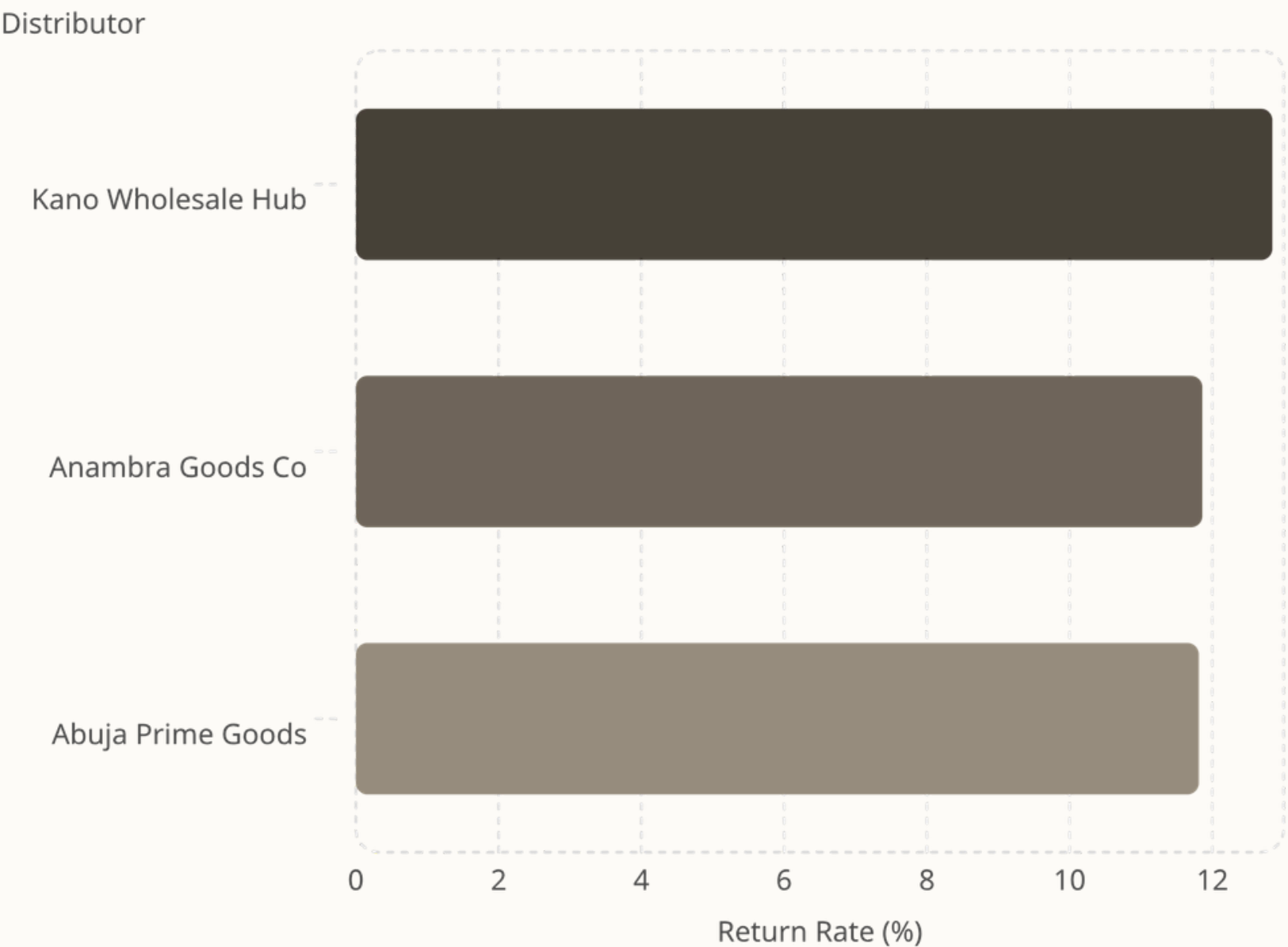
# Sales Performance Insights

Target achievement and return rate analysis identifies top performers and high-risk distributor relationships.

## Average Target Achievement by Salesperson



## Distributor Return Rates



**Kano Wholesale Hub** has the highest return rate at 12.84% — requires immediate root-cause investigation.



# Business Recommendations

## 🏆 Prioritize Top Revenue Products

Vegetable Oil and Powdered Milk drive the highest revenue — prioritize them in inventory planning, promotions, and forecasting.

## 📍 Invest in Lagos Expansion

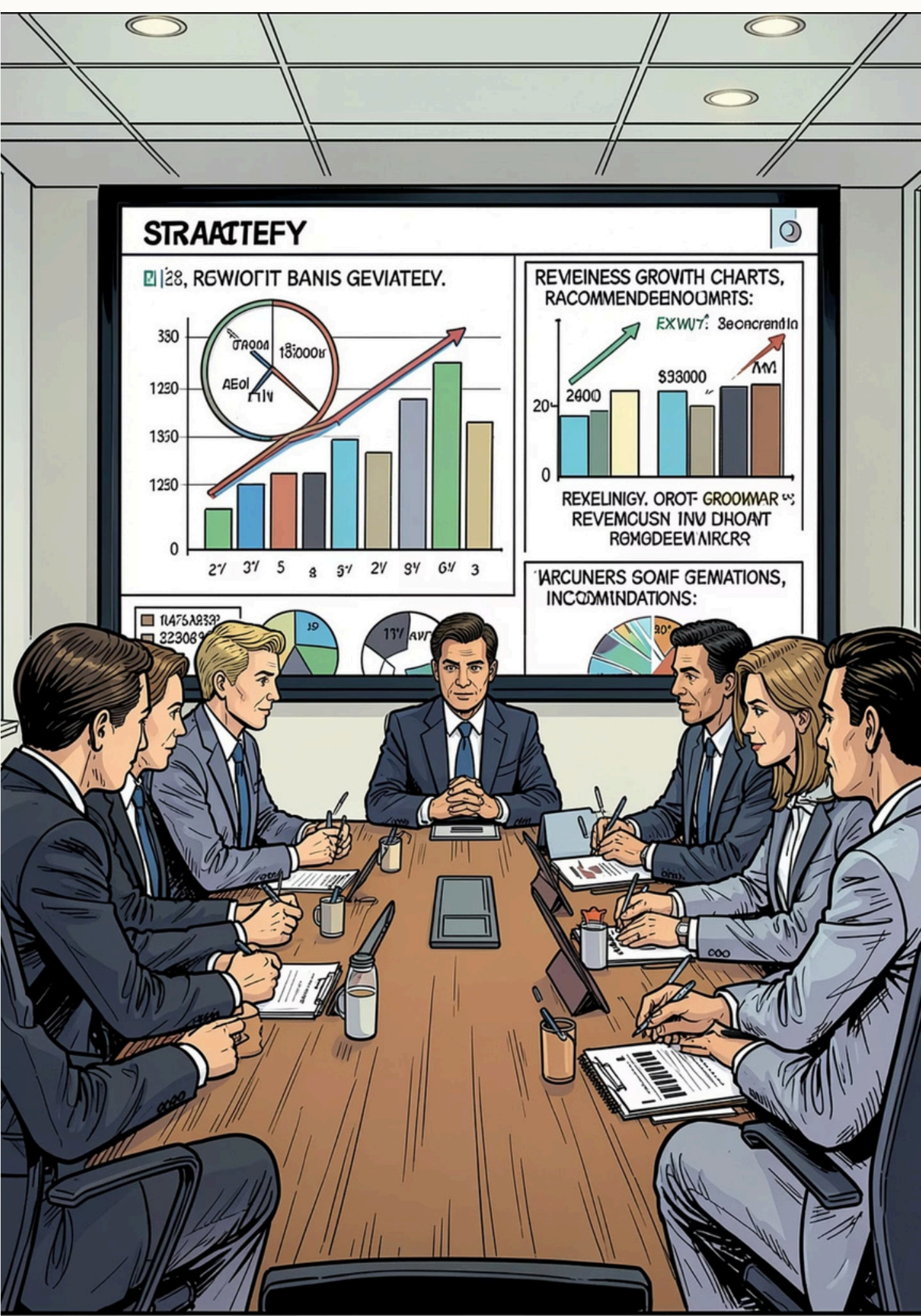
With **36.42% MoM growth**, Lagos is the strongest region for targeted commercial investment and market expansion.

## 👤 Leverage Top Performers

Salespeople exceeding 100% target should serve as benchmarks for coaching and performance improvement programs.

## 🔍 Investigate Kano Returns

Kano Wholesale Hub's 12.84% return rate demands root-cause analysis into product quality, logistics, or order fulfillment.





# Technical Achievements

A fully production-grade pipeline — built, tested, containerized, and orchestrated with industry best practices.



## Modular Python ETL

Reusable, testable extraction and transformation logic



## DuckDB Warehouse

Fast, embedded columnar analytics engine



## Airflow Orchestration

Automated DAG scheduling and dependency management



## GitHub Version Control

Full code history, branching, and collaboration workflow



## Data Validation Framework

Seven validation rules enforced at ingestion



## dbt Transformations

Staged, tested, and documented model layers



## Docker Containerization

Reproducible, portable environment for all components



## Business SQL Analytics

Actionable insights generated directly from mart models

# Key Outcomes

## ✓ Automated Pipeline

End-to-end ETL running without manual intervention

## ✓ Trusted Data Quality

Validation framework eliminates bad data at source

## ✓ Analytics Warehouse

Reliable, modeled data ready for self-service BI

## ✓ Actionable Insights

SQL-driven findings directly informing business decisions

## ✓ Scalable Architecture

Containerized design ready for cloud deployment

---

# Thank You

Questions welcome — happy to walk through the architecture, code, or business insights in more detail.